

Conceptual Answers – OFI Task

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1. What is the motivation behind measuring OFI at multiple depth levels of the order book?

Looking at OFI only at the best bid and ask (with depth 0) gives you a snapshot of the most visible liquidity, but it does not always tell the full story. A lot of meaningful order flow happens just below the surface, especially at levels 1 through 5, and ignoring those can cause us to miss important context.

By including multiple depths, we can better understand how much buying or selling pressure is really in the book. It is also useful when depth 0 is thin or is shifting quickly. I found that by weighting shallower levels more heavily, we can still prioritize the most actionable information while acknowledging that deeper liquidity matters, too.

So in short, multi-level OFI gives us a richer, more stable signal that reflects actual market dynamics better than just looking at the top level.

2. Why use Lasso regression instead of OLS for estimating cross-impact?

Cross-impact involves a lot of variables, basically, you are looking at how order flow in other assets affects the one you are focused on. If you use OLS, you are including everything and that can make the model noisy or unstable, especially when the number of predictors gets large.

Lasso is great here because it helps with both variable selection and regularization. It shrinks the coefficients of the unimportant variables to zero, so you end up with a model that focuses only on the most relevant assets. It also reduces overfitting, which is a real issue in high-dimensional problems like this.

In short, Lasso is a cleaner way to find out which assets actually matter for cross-impact without making the model overcomplicated.

3. Why is OFI a better predictor of short-term returns than trade volume?

Trade volume tells you how much activity occurred, but it does not really say who was in control: buyers or sellers. OFI, on the other hand, captures the net pressure by looking at how much size is being added or removed from bids and asks. It is directional.

In the short term, price movements are much more sensitive to the imbalance between buying and selling pressure, not just how many trades took place. So OFI gives you a more meaningful signal when trying to predict micro-price changes.

Plus, volume can spike for a lot of reasons that do not move price (like two-sided interest), but if OFI is heavily positive or negative, there is usually real directional intent behind it.